HACK APERTUS 2026 · TRACK 2A · CHALLENGE BY OST

GeaNaVid

Every claim checked against the official voting booklet, with the line that decides it.

Gea · Na · Vid — Romansh for Yes · No · Blank



▶ Demo video [TBD]

Built on Apertus v1.5

Team Apertool · solo builder

THE PROBLEM

Voters hear the claim. The answer is on page 4 of 40.

THE CLAIM

„... die Bevölkerung seit Einführung der Personenfreizügigkeit im Jahr 2002 um rund 500'000 Personen gewachsen ist.“

“... the population has grown by about 500,000 since free movement began in 2002.”

OFFICIAL BOOKLET · PAGE 4

„Seit Einführung der Personenfreizügigkeit im Jahr 2002 ist die Bevölkerung um rund 1,7 Millionen Personen gewachsen.“

Na

CONTRADICTION

Claim–label pair from the OST demo set (MNLIoverSwissVotingBooklets, label 2). Booklet: Federal Council explanations for the vote of 14 June 2026, German edition, 40 pages, PDF page 4.

THE TASK

Booklet + claim in, a ballot mark out.

Voting booklet
PDF · German, French or Italian

Claim
one sentence · DE, FR or IT

→



GeaNaVid
CLI · Apertus v1.5

→

Gea	entailment	0
Na	contradiction	2
Vid	neutral	1

+ the passages it rests on · tokens used · seconds taken

JUDGED ON

Macro-F1 on a held-out benchmark · **evidence quality** · **input tokens** · **inference time**

All 9 claim/booklet language combinations. Labels 0/1/2 and judging criteria as set by OST in the Hack Apertus challenge brief.

THE NAME

Gea. Na. Vid.

Romansh, the fourth national language, for Yes, No and Blank. All three words come from the Romansh voting booklets.

Gea

Yes → `entailment`

„... recumondan cussegl federal e parlament da metter in **Gea** *en l'urna ...“*

“... the Federal Council and Parliament recommend putting a Yes in the ballot box ...”

`Booklet of 12 June 1994`

Na

No → `contradiction`

„Nus recumandain in **NA** *a las chapellinas blaus.“*

“We recommend a NO to the blue-helmet troops.” (referendum committee)

`Booklet of 12 June 1994`

Vid

Blank, empty → `neutral`

„... abitaziuns da vacanzas ch'èn savens **vidas** *...“*

“... holiday homes that mostly stand empty ...”

`Booklet of 25 June 1995`

Romansh text from the aligned corpus VotingBooklets-v1 (Stüssi & Vamvas, SwissText 2026, CC BY 4.0); English glosses ours.

THE CONTEXT LADDER

Start with a page. Climb only when unsure.

1 **Top-5 passages**
keyword search, in the claim's own language

≈ 1,700 tokens

UNSURE? low label probability · a verdict without a cited passage

2 **The pages around the hits**
one more look, one page before and after

+ ≈ 4,100 tokens

full **The whole booklet**
baseline OST asks for

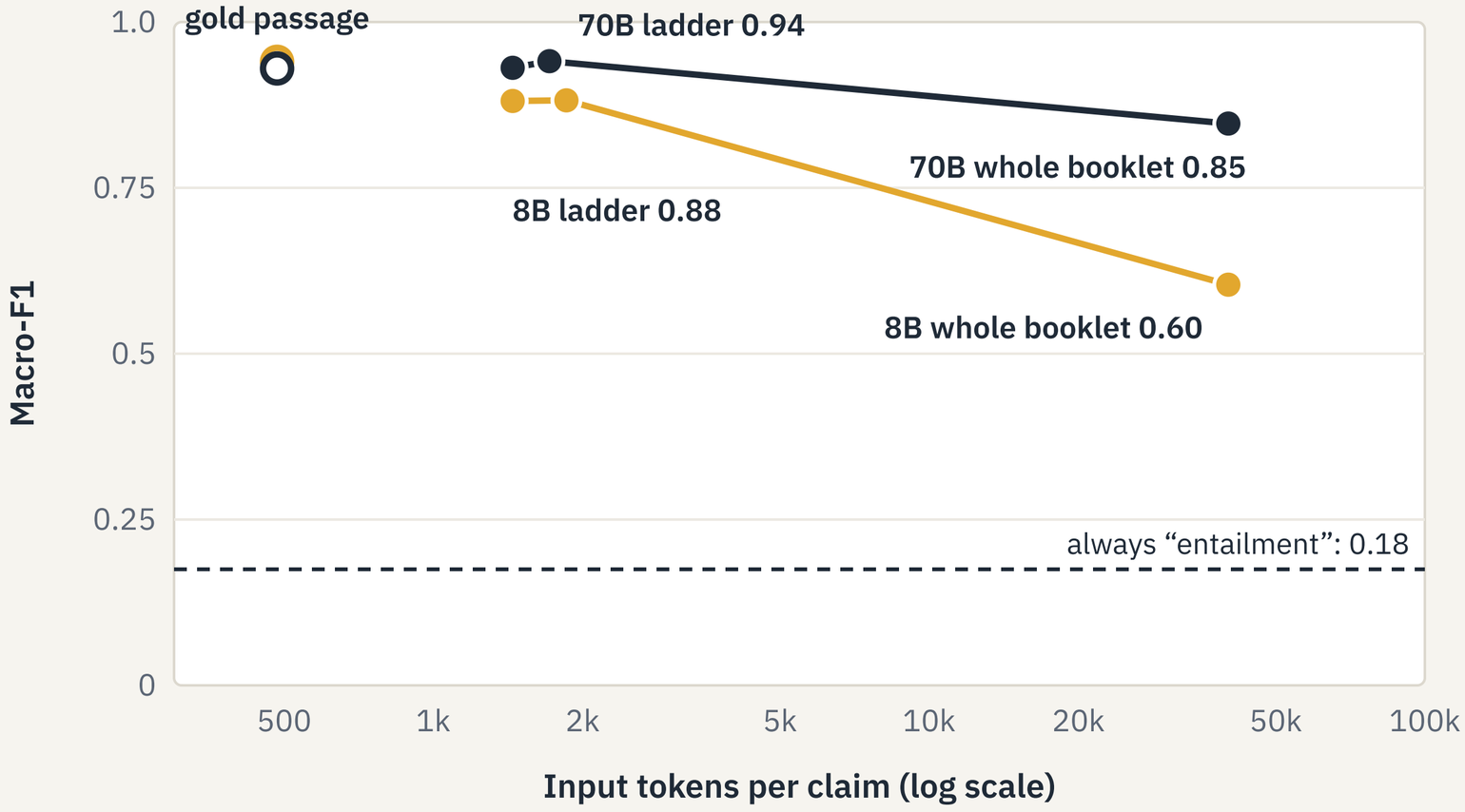
≈ 18,200 tokens

Claims settled on rung 1: 89 % (66 of 74)

Prompt tokens per claim as reported by the API, Apertus v1.5 8B, OST demo set and its cross-lingual variants (74 pairs), 1 October 2026.

THE TRADE-OFF

4 % of the tokens. Better verdicts.



Apertus 8B

Ladder 0.88 with 1,856 tokens. Whole booklet 0.60 with 40,149.

Apertus 70B

Ladder 0.94 with 1,715 tokens. Whole booklet 0.85, and 2.8 s instead of 0.9 s.

Gold passage

0.94 and 0.93. Retrieval loses 0.06 with 8B and nothing with 70B.

Held-out synthetic test: 101 claims on four booklets (2010, 2016, 2020, 2025), labels by two other LLMs with cross-review, not yet checked by a human. 95 % intervals ±0.05–0.10. Measured 1 October 2026.

ACROSS LANGUAGES

Claim in German, booklet in French. The evidence stays French.

	DE	FR	IT
DE	[TBD]	[TBD]	[TBD]
FR	[TBD]	[TBD]	[TBD]
IT	[TBD]	[TBD]	[TBD]

Macro-F1 per pair. Rows: claim language · columns: booklet language. Framed: same language.

31/48

cross-lingual demo pairs where plain keyword search finds the right page in its top 5. Without help, 17 of 48 miss.

> So GeaNaVid searches the booklet edition in the claim's language and maps the hits back by page. The right passage is then in the top 5 for 36 of 38 claims, with zero extra tokens.

Own probe: BM25 over the three editions of the 14 June 2026 booklet, 24 non-neutral demo claims × 2 other languages. Same-language recall@3 on the same claims: 22 / 24.

EVIDENCE

Every verdict points to a line in the booklet.



Quotes, not paraphrase

Evidence is a real text block from the PDF, with page and position. The model picks it; it does not write it.

Evidence first, label second

Apertus names the passages before it gives the verdict, in strict JSON.

Not voting advice

GeaNaVid says what the booklet says. It never says how to vote.

Output format as designed for the CLI; measured values follow from the benchmark runs.

TEAM APERTOOL

Gea, Na or Vid.
Always with the line that decides it.



Niklas Wichter

Team Apertool · solo builder

REPO [TBD]

VIDEO [TBD]

REPORT [TBD]

JUDGES START HERE

```
git clone [TBD]
cp .env.example .env
# set the three LLM_* variables
make run
```


Hack Apertus 2026, Track 2A, challenge by OST. GeaNaVid is an independent entry, not a product of OST or the Federal Chancellery.